

RANGEDALA, Sweden

WMO: 025360

Lat: 57.7846N

Lon: 13.1642E

Elev: 981

StdP: 14.18

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.1	12.2	4.3	7.1	11.4	8.2	8.7	15.8	19.5	32.3	17.8	33.2	2.5	20	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	77.2	62.4	74.2	61.4	71.0	60.0	65.2	72.5	63.4	70.7	61.7	68.4	5.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.7	88.4	67.9	60.7	82.2	65.9	58.9	77.1	63.7	30.7	72.2	29.2	71.1	28.0	68.4	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.7	13.8	12.1	3.1	81.8	5.1	2.9	-0.6	83.9	-3.5	85.7	-6.4	87.3	-10.1	89.5		
(5)			2.5	68.7	4.8	2.6	-1.0	70.6	-3.8	72.1	-6.5	73.6	-10.0	75.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	27.4	28.4	33.1	41.9	50.0	55.9	60.3	59.1	52.2	43.6	36.0	30.8		
	DBStd	13.29	7.77	7.47	7.01	6.82	6.79	5.56	5.20	4.91	4.95	6.23	6.66	7.93		
	HDD50	3439	700	604	524	257	84	8	0	2	34	210	420	596		
	HDD65	7956	1165	1024	989	692	467	280	165	194	385	664	870	1061		
	CDD50	996	0	0	0	15	84	184	320	283	99	11	0	0		
	CDD65	37	0	0	0	0	1	6	20	10	0	0	0	0		
	CDH74	279	0	0	0	1	13	62	156	47	0	0	0	0		
	CDH80	23	0	0	0	0	0	3	18	2	0	0	0	0		
	WSAvg	5.8	6.2	6.5	6.5	6.2	5.7	4.9	4.4	4.6	5.1	6.0	6.7	6.4		
	PrecAvg	37.1	3.4	2.4	2.0	2.2	2.4	3.3	3.4	3.5	3.4	3.6	3.4	3.6		
(15) Precipitation	PrecMax	45.5	7.2	6.8	4.0	4.8	4.6	5.9	7.1	6.9	8.0	8.1	6.1	7.6		
	PrecMin	25.6	0.4	0.6	0.6	0.2	0.1	0.0	0.5	1.3	0.8	0.0	1.3	0.6		
	PrecStd	5.5	1.8	1.5	0.9	1.1	1.0	1.4	1.6	1.5	2.0	1.8	1.2	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.1	44.5	55.4	67.2	75.8	79.8	82.4	77.9	70.1	59.4	51.0	47.0		
		MCWB	42.6	41.9	44.7	53.6	59.0	62.1	64.8	63.9	60.4	54.4	49.9	46.4		
	2%	DB	40.9	41.6	49.9	62.3	71.0	75.3	78.2	75.0	66.0	55.5	48.6	44.5		
		MCWB	40.4	40.3	42.7	50.1	57.0	59.9	63.3	63.2	58.4	53.8	47.7	44.0		
	5%	DB	39.0	39.8	45.1	57.7	66.9	71.3	74.9	71.6	62.5	53.4	46.8	42.8		
		MCWB	38.4	38.7	40.6	48.0	54.5	58.5	62.3	61.7	56.5	51.7	46.0	42.1		
	10%	DB	37.3	38.0	41.8	53.1	62.4	67.0	71.2	67.9	59.6	51.7	44.9	40.8		
		MCWB	36.7	36.9	38.6	45.3	52.5	56.6	61.1	60.5	55.4	50.2	44.2	40.2		
	0.4%	WB	42.9	43.1	46.4	55.7	61.9	65.1	67.8	67.2	61.9	56.1	50.4	46.6		
		MCDB	43.1	43.8	52.9	65.4	72.0	74.6	75.9	73.1	67.1	57.6	50.9	46.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.6	41.0	43.7	51.6	58.7	61.8	65.7	64.8	59.3	54.0	48.0	44.1		
		MCDB	40.7	41.4	48.3	59.5	67.2	71.3	73.5	70.7	64.0	55.1	48.4	44.4		
	5%	WB	38.6	38.9	41.2	48.7	56.2	59.7	64.0	63.0	57.5	52.2	46.1	42.3		
		MCDB	38.9	39.5	44.2	55.4	63.8	68.1	71.9	68.9	60.9	53.2	46.6	42.6		
	10%	WB	36.8	37.1	39.2	46.4	53.8	57.8	62.1	61.3	56.0	50.5	44.3	40.2		
		MCDB	37.2	37.8	41.2	52.0	61.1	65.3	69.4	66.8	58.9	51.5	44.9	40.7		
	5% DB	MDBR	6.9	7.4	10.8	14.3	15.7	15.1	14.6	13.1	11.7	8.7	6.3	6.4		
		MCDBR	7.1	8.1	16.1	20.8	21.8	21.7	21.5	18.6	15.4	9.4	7.4	7.1		
		MCWBR	7.1	6.9	10.3	11.6	11.5	10.5	9.8	9.0	8.9	7.3	7.2	7.4		
		MCWBR	7.1	7.0	13.9	18.2	19.5	19.0	17.8	15.3	12.9	8.7	6.9	7.0		
(35) Mean Daily Temperature Range	5% WB	MCWBR	7.3	6.6	9.7	10.7	11.2	10.2	9.4	8.7	8.3	7.5	6.9	7.3		
	taub		0.281	0.312	0.329	0.367	0.364	0.354	0.375	0.372	0.342	0.331	0.302	0.266		
		taud	2.276	2.347	2.379	2.335	2.400	2.447	2.402	2.414	2.489	2.442	2.334	2.210		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		185	232	261	268	276	281	271	264	256	223	174	150		
		Edh at Noon	15	22	28	34	34	33	34	31	25	20	14	12		
	RadAvg		116	308	765	1232	1617	1768	1609	1269	878	403	149	81		
(45) All-Sky Solar Radiation	RadStd		23	49	76	166	148	115	211	123	108	67	28	10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page